

Some Background Notes

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In the following, the numbers in parentheses refer to the relevant page number in Kreyszig's functional analysis text [2].

1 Functional Analysis Review

1.1 Linear Operators

Let $L(X, Y)$ denote the set of linear operators from the vector space X to the vector space Y . A linear operator is defined as follows:

Definition 1.1 (Linear Operator (82, 83)). *Let X and Y be vector spaces over the same field. Let $D(T) \subset X$. An operator $T : D(T) \rightarrow Y$ is called linear if it satisfies $T(ax + by) = aTx + bTy$ for all $x, y \in D(T)$ and for any scalars a, b .*

Assuming that $T \in L(X, Y)$ is injective, there exists a mapping $T^{-1} : R(T) \rightarrow D(T)$ that sends every $y \in R(T)$ to the unique $x \in D(T)$ for which $Tx = y$. Some useful properties of T^{-1} are given in the following theorem:

Theorem 1.2 (Inverse Operator (88)). *Let X and Y be vector spaces over the same field. Let $T : D(T) \rightarrow Y$ be a linear operator with $D(T) \subset X$ and $R(T) \subset Y$. Then*

1. $T^{-1} : R(T) \rightarrow D(T)$ exists if and only if T is injective.
2. If T^{-1} exists it is a linear operator.
3. If $\dim(D(T)) = n < \infty$ and T^{-1} exists then $\dim(R(T)) = \dim(D(T))$.

Bounded, compact, and closed linear operators will be used throughout. A bounded linear operator is defined as follows:

Definition 1.3 (Bounded Linear Operator (91)). Let X and Y be normed spaces. $T \in L(X, Y)$ is a bounded linear operator if there exists a $c > 0$ such that $\|Tx\|_Y \leq c\|x\|_X$ for all $x \in D(T)$.

We use $B(X, Y)$ to denote the set of all bounded linear operators from X to Y , equipped with the norm

$$\|T\|_{B(X, Y)} = \sup_{x \in X} \frac{\|Tx\|_Y}{\|x\|_X}. \quad (1)$$

Similarly, we will use $K(X, Y)$ to denote the set of compact linear operators from X to Y . A compact linear operator is defined as follows:

Definition 1.4 (Compact Linear Operator (406, 407)). Let X and Y be normed spaces. $T \in L(X, Y)$ is a compact linear operator if for every bounded subset M of X , the image $T(M)$ is relatively compact, that is, the closure $\overline{T(M)}$ is compact. Equivalently, T is compact if and only if it maps every bounded sequence $\{x_n\}_{n \in \mathbb{N}}$ in X onto a sequence $\{Tx_n\}_{n \in \mathbb{N}}$ in Y that has a convergent subsequence.

Every compact linear operator is bounded. Thus, $K(X, Y) \subset B(X, Y)$ (406). The following theorem describes the situation in which the composition of a bounded linear operator with a compact operator is compact [4].

Theorem 1.5. Consider $S \in B(X, X)$ and $T \in K(X, Y)$. The operator TS is compact, i.e., $TS \in K(X, Y)$.

Proof. Let $\{x_n\}_{n \in \mathbb{N}}$ be a bounded sequence in X . Since $S \in B(X, Y)$, $\|Sx_n\| \leq \|S\|_{B(X, Y)}\|x_n\|_X$ for all $n \in \mathbb{N}$. Since $\{x_n\}_{n \in \mathbb{N}}$ is a bounded sequence and S is a bounded operator, we can say $\|Sx_n\| < \infty$. Thus, $\{Sx_n\}_{n \in \mathbb{N}}$ is a bounded sequence in X . Since T is compact, applying T to a bounded sequence in X yields a sequence in Y with a convergent subsequence. Thus, $T\{Sx_n\}_{n \in \mathbb{N}} = (TS)\{x_n\}_{n \in \mathbb{N}}$ has a convergent sequence in Y and $TS \in K(X, Y)$. $\square$

In spectral theory, we will often consider closed operators, which are defined as follows:

Definition 1.6 (Closed linear operator (292)). Let X and Y be normed spaces. $T \in L(X, Y)$ is called a closed linear operator if its graph,

$$G(T) = \{(x, y) \mid x \in D(T), y = Tx\}, \quad (2)$$

is closed in the normed space $X \times Y$, where the two algebraic operations of the vector space $X \times Y$ are defined as usual, and the norm is defined by $\|(x, y)\|_{X \times Y} = \|x\|_X + \|y\|_Y$.

Alternatively, closed operators can be defined using the following theorem:

Theorem 1.7 (Closed linear operator (293)). *Let X and Y be normed spaces. $T \in L(X, Y)$ is a closed linear operator if and only if it has the following property: whenever $\{x_n\}_{n \in \mathbb{N}}$ is a sequence in $D(T)$ such that $x_n \rightarrow x$ and $Tx_n \rightarrow y$, then $x \in D(T)$ and $Tx = y$.*

Going forward, $C(X, Y)$ will denote the set of closed linear operators from X to Y . To close out this section, we recall the Closed Graph Theorem:

Theorem 1.8 (Closed Graph Theorem (292)). *Let X and Y be Banach spaces and $T \in C(X, Y)$ with $D(T) \subset X$. If $D(T)$ is closed in X , then T is bounded.*

1.2 A Survey of Results

Next, we collect several important results that will be useful in the proofs that follow. We first consider the Open Mapping Theorem and the Bounded Inverse Theorem, whose proofs are given in [2].

Definition 1.9 (Open mapping (286)). *Let X and Y be two metric spaces. Then $T : D(T) \rightarrow Y$, where $D(T) \subset X$, is called an open mapping if for every open set in $D(T)$, the image is an open set in Y .*

Theorem 1.10 (Open Mapping Theorem (286)). *Let X and Y be two Banach spaces, i.e., complete normed spaces. $T \in B(X, Y)$ is an open mapping.*

Theorem 1.11 (Bounded Inverse Theorem (286)). *Let X and Y be two Banach spaces. If $T \in B(X, Y)$ is bijective, then T^{-1} is bounded.*

Often, we will make use of the Neumann series, defined in the following theorem:

Theorem 1.12 (Neumann Series (375)). *Let $T \in B(X, X)$, where X is a Banach space. If $\|T\|_{B(X, X)} < 1$, then $(I - T)^{-1}$ exists as a bounded linear operator on the whole space X . Furthermore,*

$$(I - T)^{-1} = \sum_{j=1}^{\infty} T^j = I + T + T^2 + \dots, \quad (3)$$

where the series on the right is convergent in the norm on $B(X, X)$.

Finally, we recall two results that will be useful in understanding the spectra of compact operators.

Theorem 1.13 (Riesz's Lemma [4]). *Let W and X be normed spaces where W is a closed proper subset of X . Then for every $\epsilon > 0$, there exists a $x \in X$ such that $\|x\|_X = 1$ and $\text{dist}(x, W) \geq 1 - \epsilon$.*

Corollary 1.14 (Riesz's Ball Theorem (80)). *Let X be a normed linear space and B be the closed unit ball in X . If X is finite dimensional then B is compact. If X is infinite-dimensional, then B is never compact.*

1.3 Banach and Hilbert Adjoints

Recall that a linear functional is a linear operator with its domain in a vector space and its range in the corresponding scalar field. Similarly, a linear functional is bounded when it is a bounded linear operator.

Before defining the Adjoint of a linear operator, we first recall the definition of a dual space.

Definition 1.15 (dual space (119)). *Let X be a normed space. The set of all bounded linear functionals on X is the normed space X^* , referred to as the dual space of X . The dual space is equipped with the following norm:*

$$\|f\| = \sup_{\substack{x \in X \\ \|x\|_X=1}} |f(x)|. \quad (4)$$

Let X and Y be two Banach spaces and let $T \in B(X, Y)$. We use $\langle x_*, x \rangle_{X^*, X}$ to denote the action of the element $x_* \in X^*$ on an element $x \in X$. In other words, $\langle x_*, x \rangle_{X^*, X}$ denotes the application of the functional x_* to x .

For any $y^* \in Y^*$ and $x \in X$, define the functional $G_{y^*}(x) = \langle y^*, Tx \rangle_{Y^*, Y}$. From the linearity and boundedness of T and y^* , we find that G_{y^*} is a bounded linear functional on X , and is thus an element of X^* . Using this functional, we define the Banach adjoint as follows:

Definition 1.16 (Banach Adjoint [4]). *Let X and Y be Banach spaces and $T \in B(X, Y)$. The Banach adjoint of T , denoted T^* , is the operator in $B(Y^*, X^*)$ that maps $y^* \in Y^*$ to G_{y^*} , where $G_{y^*}(x) = \langle y^*, Tx \rangle$.*

Let H denote a Hilbert space, i.e., a complete inner product space, with norm $\|\cdot\|_H$ and inner product $(\cdot, \cdot)_H$. To define the Hilbert adjoint, we first recall the following result:

Theorem 1.17 (Riesz Representation Theorem (188)). *Given any $h_* \in H^*$, there exists a unique $h \in H$ such that $\langle h_*, k \rangle_{H^*, H} = (h, k)_H$ for all $k \in H$. Furthermore, $\|h\|_H = \|h\|_{H^*}$.*

Using this result, we define the Riesz map of H as follows:

Definition 1.18 (Riesz map [4]). *Following the notation in Theorem 1.17, we define the Riesz map of H as the map $R_H : H^* \rightarrow H$ that maps h_* to h . This map is*

bijective.

Using the Riesz map, the Hilbert adjoint¹ is defined as follows:

Definition 1.19 (Hilbert Adjoint [4]). Let H_1 and H_2 be two Hilbert spaces. Consider $T \in B(H_1, H_2)$ and its Banach adjoint $T^* \in B(H_2^*, H_1^*)$. The Hilbert adjoint is the operator $T' : H_2 \rightarrow H_1$ that maps $x \in H_2$ to $R_{H_1} T^* R_{H_2}^{-1} x \in H_1$, where R_{H_1} and R_{H_2} are the Riesz maps for H_1 and H_2 , respectively.

Alternatively, we can define the Hilbert adjoint with the following result

Theorem 1.20 (Hilbert Adjoint [4]). Let H_1 and H_2 be two Hilbert spaces and $T \in B(H_1, H_2)$. The Hilbert adjoint, $T' : H_2 \rightarrow H_1$ can be defined as the operator in $B(H_2, H_1)$ that satisfies $(Th, x)_{H_2} = (h, T'x)_{H_1}$ for all (h, x) in $H_1 \times H_2$.

Proof. Using Definition 1.19,

$$(h, T'x)_{H_1} = (h, R_{H_1} T^* R_{H_2}^{-1} x)_{H_1}. \quad (5)$$

Using the definition of the Riesz map R_{H_1} ,

$$\begin{aligned} (h, T'x)_{H_1} &= (h, R_{H_1} T^* R_{H_2}^{-1} x)_{H_1} \\ &= \langle T^* R_{H_2}^{-1} x, h \rangle_{H_1^*, H_1}. \end{aligned} \quad (6)$$

Using the definition of the Banach adjoint in Definition 1.16, we have

$$\begin{aligned} (h, T'x)_{H_1} &= (h, R_{H_1} T^* R_{H_2}^{-1} x)_{H_1} \\ &= \langle T^* R_{H_2}^{-1} x, h \rangle_{H_1^*, H_1} \\ &= \langle R_{H_2}^{-1} x, Th \rangle_{H_2^*, H_2}. \end{aligned} \quad (7)$$

Finally, using the definition of the Riesz map R_{H_2} we obtain the desired result:

$$\begin{aligned} (h, T'x)_{H_1} &= (h, R_{H_1} T^* R_{H_2}^{-1} x)_{H_1} \\ &= \langle T^* R_{H_2}^{-1} x, h \rangle_{H_1^*, H_1} \\ &= \langle R_{H_2}^{-1} x, Th \rangle_{H_2^*, H_2} \\ &= (Th, x)_{H_2}. \end{aligned} \quad (8)$$

□

We now list some general properties of the Hilbert adjoint that are used frequently in applications. The proofs of these results can be found in [2].

¹To compare the Banach and the Hilbert adjoints, consider the examples presented in Umberto's first lecture [4]. These are worked out in detail in my written notes.

Theorem 1.21 (Properties of the Hilbert adjoint (199)). *Let H_1 and H_2 be two Hilbert spaces. Consider $S \in B(H_1, H_2)$ and $T \in B(H_1, H_2)$. Then for any scalar α we have*

$$(a) \text{ For any } x \in H_1 \text{ and } y \in H_2, (T'y, x)_{H_1} = (y, Tx)_{H_2}$$

$$(b) (S + T)' = S' + T'$$

$$(c) (\alpha T)' = \bar{\alpha} T'$$

$$(d) (T')' = T$$

$$(e) \|T'T\|_{B(H_1, H_1)} = \|TT'\|_{B(H_2, H_2)} = \|T\|_{B(H_1, H_2)}^2$$

$$(f) T'T = 0 \iff T = 0$$

$$(g) \text{ When } H_1 = H_2 \text{ we have } (ST)' = T'S'.$$

1.4 Self-Adjoint, Unitary, and Normal Operators

Let H be a Hilbert space, and let $T \in B(H, H)$ be an endomorphism. T is self-adjoint if $T' = T$. Furthermore, if $T \in B(H, H)$ is self-adjoint and invertible, $T^{-1} \in B(H, H)$ is self-adjoint as well [2].

If $T \in B(H, H)$ satisfies $T'T = TT'$, it is called normal. Notice that when T is self-adjoint, it is normal. However, a normal operator need not be self-adjoint. $T \in B(H, H)$ is said to be unitary if T is bijective and $T' = T^{-1}$. If T is unitary, then T is normal. As before, the converse is generally not true [2].

We describe some criteria for self-adjointness in the following theorem:

Theorem 1.22 (self-adjointness (203)). *Let H be a Hilbert space and $T \in B(H, H)$. Then*

$$(a) \text{ If } T \text{ is self-adjoint, } (Tx, x)_H \text{ is real for all } x \in H.$$

$$(b) \text{ If } H \text{ is complex and } (Tx, x)_H \text{ is real for all } x \in H, \text{ then } T \text{ is self-adjoint.}$$

1.5 Beginner Spectral Theory

In this section, we review some basic concepts from spectral theory, one of the main branches of modern functional analysis. We begin with a review of spectral theory in finite dimensions.

1.5.1 Spectral Theory in Finite Dimensions

Let W be an n -dimensional normed space, where $n < \infty$. Let $T \in L(W, W)$ be represented by the matrix $A \in \mathbb{C}^{n \times n}$.

Definition 1.23 (Eigenvalues, eigenvectors, eigenspaces, spectrum, resolvent set (364)). Consider the following equation

$$Ax = \lambda x. \quad (9)$$

An eigenvalue of A is a $\lambda \in \mathbb{C}$ such that (9) has a nontrivial solution, $x \neq 0$. This nontrivial solution is called an eigenvector of A corresponding to λ . The set of all eigenvectors corresponding to λ , together with the zero vector, forms a vector subspace of W called the eigenspace of A corresponding to λ . The set of all the eigenvalues of A is called the spectrum of A , denoted $\Gamma(A)$. Its complement is called the resolvent set, denoted $\rho(A) = \mathbb{C} \setminus \Gamma(A)$.

To find the eigenvalues and corresponding eigenvectors of A , we rewrite (9) as

$$(A - \lambda I)x = 0, \quad (10)$$

where I is the identity matrix in $\mathbb{C}^{n \times n}$. For this equation to have a nonzero solution, we require that the determinant of $(A - \lambda I)$ is zero. Thus, the eigenvalues of A can be found by solving $\det(A - \lambda I) = 0$, which is called the characteristic polynomial of A . Once the eigenvalues have been computed, the corresponding eigenvectors are found by solving (10).

Each choice of basis for W corresponds to a different matrix $A \in \mathbb{C}^{n \times n}$ used to represent $T \in L(W, W)$. Although the matrix representation of T is not unique, all choices of A have the same eigenvalues (366). Thus, the eigenvalues of T are the eigenvalues of A . Similarly, the spectrum and resolvent set of A are the spectrum and resolvent set of T .

1.5.2 Spectral Theory in Infinite Dimensions

Let $X \neq \{0\}$ be a normed space and $T \in L(X, X)$. With T we associate the operator $T_z = T - zI$, where $z \in \mathbb{C}$ and I is the identity operator on $D(T) \subset W$. If T_z has an inverse, we call it the resolvent operator of T , denoted $R_T(z) = T_z^{-1} = (T - zI)^{-1}$.

To dive deeper into spectral theory, we will need to define its most fundamental concepts, namely, the regular values of T , the resolvent set of T , and the spectrum of T .

Definition 1.24 (Regular value (371)). Let $T \in L(X, X)$. A regular value of T is a $z \in \mathbb{C}$ such that

(R1) $R_T(z)$ exists,

(R2) $R_T(z)$ is bounded,

(R3) $R_T(z)$ is defined on a set that is dense in X .

Using this definition, we can define the resolvent set as follows

Definition 1.25 (Resolvent set and Spectrum (371)). The resolvent set of T , denoted $\rho(T)$, is the set of all regular values of T . The spectrum of T , denoted $\sigma(T)$, is the complement of $\rho(T)$, i.e., $\sigma(T) = \mathbb{C} \setminus \rho(T)$.

The spectrum of T can be partitioned into three disjoint subsets. We define these subsets in turn.

Definition 1.26 (Point spectrum (371)). The point spectrum, denoted $\sigma_p(T)$, is the set of $z \in \mathbb{C}$ for which $R_T(z)$ does not exist. Thus, if $z \in \sigma_p(T)$, then T does not satisfy (R1) in Definition 1.24.

Definition 1.27 (Continuous Spectrum (371)). The continuous spectrum, denoted $\sigma_c(T)$, is the set of $z \in \mathbb{C}$ for which $R_T(z)$ is defined on a set that is dense in X , but is not bounded. Thus, if $z \in \sigma_c(T)$, then T satisfies (R1) and (R3) but not (R2) in Definition 1.24.

Definition 1.28 (Residual Spectrum (371)). The residual spectrum, denoted $\sigma_r(T)$, is the set of $z \in \mathbb{C}$ for which $R_T(z)$ exists but is not defined on a set that is dense in X . Thus, if $z \in \sigma_r(T)$, then T satisfies (R1) but not (R3) in Definition 1.24. Here, T may or may not satisfy (R2).

In Table 1, we use Definition 1.24 to compare the definitions of each of these subsets. We note that the four sets in Table 1 are disjoint and their union is the complex plane, i.e., $\mathbb{C} = \rho(T) \cup \sigma(T) = \rho(T) \cup \sigma_p(T) \cup \sigma_c(T) \cup \sigma_r(T)$.

From Theorem 1.2, we know that when T_z is injective, $R_T(z)$ exists and is linear. When T_z is not injective, $R_T(z)$ does not exist. In particular, if $T_z x = (T - zI)x = 0$ for some $x \neq 0$, then $R_T(z)$ does not exist and $z \in \sigma_p(T)$. In this case, z is called an eigenvalue of T . The corresponding vector x is called an eigenvector of T . The subspace of $D(T)$ consisting of the eigenvectors of T corresponding to z , together with zero, is the eigenspace of T corresponding to z .

Notice that this definition of an eigenvalue of T is in harmony with the finite-dimensional setting in Definition 1.23. Furthermore, in comparing Definition 1.23 to

Satisfied	Not satisfied	λ belongs to:
(R1), (R2), (R3)		$\rho(T)$
(R2), (R3)	(R1)	$\sigma_p(T)$
(R1), (R3)	(R2)	$\sigma_c(T)$
(R1), (R2)	(R3)	$\sigma_r(T)$

Table 1: Classifying the resolvent set and spectrum of T using Definition 1.24.

Definition 1.26, we find that the spectrum of a linear operator on a finite-dimensional space is the point spectrum, and both the continuous spectrum and the residual spectrum are empty. Thus, when X is finite-dimensional, every spectral value of T , i.e., every $z \in \sigma(T)$, is an eigenvalue. However, when X is infinite-dimensional, T can have spectral values that are not eigenvalues.

Before moving on, we record the following facts, which will be useful as we dive deeper into spectral theory.

Theorem 1.29 (Domain of $R_T(z)$ (373)). *Let X be a Banach space, $T \in L(X, X)$, and $z \in \rho(T)$. If T is closed or if T is bounded, then $R_T(z)$ is defined on the whole space X and is bounded.*

Theorem 1.30 (Fredholm Alternative for Compact Operators [4]). *Let X be a Banach space and $T \in K(X, X)$. Then $N(T - I) = \{0\}$ if and only if $R(T - I) = X$. In other words, $T - I$ is injective if and only if it is surjective. Furthermore, $N(T - I)$ is finite-dimensional.*

Corollary 1.31 (Bounded $R_T(z)$ [4]). *Let X be a Banach space and $z \in \mathbb{C} \setminus \{0\}$. Either $Tx = zx$ has nontrivial solutions or T_z is bijective and $R_T(z) \in B(X, X)$.*

Proof. Consider $T_z = T - zI = -z(I - z^{-1}T)$. Since T is compact and $z \neq 0$, $P = z^{-1}T$ is compact. By the Fredholm Alternative, $N(I - P) = \{0\} \iff R(I - P) = X$. When $N(I - P) \neq \{0\}$, then $(I - P)x = 0$, or equivalently $Tx = zx$, has nontrivial solutions. When $N(I - P) = \{0\}$, then $I - P$ is bijective. Thus, $-z(I - P) = T_z$ is bijective. Furthermore, since T is compact and thus bounded, T_z is bounded. Applying the Bounded Inverse Theorem (Theorem 1.11), we have that $R_T(z) = T_z^{-1}$ exists and is bounded. $\square$

2 An Introduction to Spectra

Let $T : D(T) \rightarrow H_2$ denote a linear operator between Hilbert spaces H_1 and H_2 . We say that T is boundedly invertible if there exists a $B \in B(H_2, H_1)$ such that $TB = I$ and $BT \subset I$ ² [1]. In this case, B is called the inverse of T .

We note that when T is a closed operator, T is boundedly invertible when $N(T) = \{0\}$ and $R(T) = H_2$. This follows from the Closed Graph Theorem, stated as Theorem 1.8 above. Furthermore, when $T \in B(H_1, H_2)$ is bijective, the Bounded Inverse Theorem (Theorem 1.11) says that $T^{-1} \in B(H_2, H_1)$. Thus, when T is bounded and bijective, invertibility is equivalent to bounded invertibility.

In this setting, the spectrum of T is defined as follows

Definition 2.1 (Spectrum and Resolvent Set of T [1]). *Let T be a linear operator defined on H . The spectrum of T , denoted $\sigma(T)$, is the following set:*

$$\sigma(T) = \{z \in \mathbb{C} \mid T_z = T - zI \text{ is not boundedly invertible}\} \quad (11)$$

The complement of $\sigma(T)$ is called the resolvent set, denoted $\rho(T) = \mathbb{C} \setminus \sigma(T)$.

We note that this definition of $\sigma(T)$ is stronger than the definition of $\sigma(T)$ in Definition 1.25. In Definition 1.25, z is a spectral value of T when either $R_T(z)$ does not exist, $R_T(z)$ is not bounded, or $D(R_T(z))$ is not dense in H . However, in Definition 2.1, z is a spectral value of T if either $R_T(z)$ does not exist or $R_T(z)$ exists and is not bounded. This difference is due to our definition of bounded invertibility, which forces the inverse of T_z to be defined on the full Hilbert space. Thus, when considering bounded invertibility, we assume that when $R_T(z)$ exists, $D(R_T(z)) = H$.

When T is closed, so is T_z . When T_z is closed and $R_T(z)$ exists and is bounded, $D(R_T(z))$ is closed³. Thus, $D(R_T(z))$ is dense in H if and only if $D(R_T(z)) = H$. In other words, when T is closed the two definitions of $\sigma(T)$ are equivalent.

Going forward, we will use Definition 2.1 as our definition of $\sigma(T)$. In this setting, if T is not closed, then $\sigma(T) = \mathbb{C}$ [1]. Thus, we will consider only closed linear operators, whose spectra may be nontrivial to compute.

² $BTx = x$ for every $x \in D(T)$. Thus, BT is only defined on $D(T)$ while I is defined on the full Hilbert space H_1 . Thus, $BT \subset I$.

³Let $y_n \rightarrow y$ be a sequence in $D(R_T(z))$ that converges to a $y \in H$. Since $y_n \in D(R_T(z)) = R(T_z)$, there exists $x_n \in D(T_z)$ such that $x_n = R_T(z)y_n$. Since $R_T(z)$ is bounded, $\|x_n - x_m\| = \|R_T(z)(y_n - y_m)\| \leq c\|y_n - y_m\|$. Since y_n is Cauchy in H , x_n is Cauchy in H . Hence, x_n converges to some $x \in H$. In particular, $x_n \rightarrow x$ and $T_z x_n = y_n \rightarrow y$. Since T_z is closed, $x \in D(T_z)$ and $T_z x = y$. Thus, $y \in R(T_z) = D(R_T(z))$ and $D(R_T(z))$ is closed.

2.1 Properties of the Spectrum

The following theorem is the basis of many of the results to come.

Theorem 2.2 (First Stability Estimate [3]). ⁴ *Let X and Y be two Banach spaces. Suppose $T \in C(X, Y)$ has a bounded inverse T^{-1} . Then for any $E \in B(X, Y)$ with $\|E\|_{B(X, Y)} < 1/\|T^{-1}\|_{B(Y, X)}$, $T + E$ has a bounded inverse $(T + E)^{-1}$ satisfying*

$$\|(T + E)^{-1}\|_{B(Y, X)} \leq \frac{\|T\|_{C(X, Y)}}{1 - \|E\|_{B(X, Y)}\|T^{-1}\|_{B(Y, X)}}. \quad (12)$$

Proof. In written notes. To be added here soon! □

Using this result, we can prove the following results:

Corollary 2.3. *Let X be a Banach space and $T \in C(X, X)$. Then $\rho(T)$ is open and $\sigma(T)$ is closed.*

Proof. In written notes. To be added here soon! □

Corollary 2.4. *Let X be a Banach space and $T \in B(X, X)$. Then $\sigma(T)$ is bounded and nonempty.*

Proof. In written notes. To be added here soon! The idea is that if $\sigma(T)$ is empty, then $R_T(z)$ is entire and bounded. Thus, Liouville's theorem tells us that $R_T(z)$ is constant. Then, showing that $R_T(z) \rightarrow 0$ and $|z| \rightarrow \infty$, we have that $R_T(z) = 0$ which is a contradiction. □

2.2 Eigenvalues

As before, $z \in \mathbb{C}$ is an eigenvalue of T if $T_z = T - zI$ is not injective, i.e., $N(T_z) \neq \{0\}$. We use $\Gamma(T)$ to denote the set of eigenvalue of T . For each $z \in \Gamma(T)$, the corresponding null space of T_z , $N(T_z)$, is the eigenspace of T corresponding to z . Notice that $\Gamma(T) \subset \sigma(T)$. However, there may exist $z \in \sigma(T) \setminus \Gamma(T)$. For example, consider the right shift operator $T_r : l^2 \rightarrow l^2$ that maps $(u_1, u_2, \dots)$ to $(0, u_1, u_2, \dots)$. The only possible eigenvector of T_r is the zero vector. Thus, $\Gamma(T_r) = \emptyset$. However, since T_r is a bounded operator, $\sigma(T_r)$ is nonempty⁵.

⁴I need to compare Theorem 4.2 in [3] to Umberto's result in [4] to decide if $T \in C(X, Y)$, $T \in C(X, X)$, or $T \in B(X, Y)$.

⁵We have omitted a discussion of the ascent of a nonzero eigenvalue of a compact operator T . This can be found in my written notes.

2.3 The Norm of the Resolvent

Much of our work will involve the norm of the resolvent, that is, $\|R_T(z)\|$. Some of its important properties are summarized in the following results:

Theorem 2.5 (Norm of the resolvent [3]). *Let $T \in C(X, X)$, where X is a Banach space. For $z \in \rho(T)$, $\|R_T(z)\|_X$ satisfies the following bound:*

$$\|R_T(z)\|_X \geq \frac{1}{\text{dist}(z, \sigma(T))}, \quad (13)$$

where $\text{dist}(z, \sigma(T)) = \inf_{w \in \sigma(T)} |z - w|$.

Proof. In written notes. To be added here soon! □

Theorem 2.6 (Upper-semicontinuity [3]). *Let $T \in C(X, X)$, where X is a Banach space. If $z \notin \sigma(T)$, then $z \notin \sigma(T + E)$ for any $E \in B(X, X)$ that satisfies $\|E\|_X \leq 1/\|R_T(z)\|_X$.*

Proof. In written notes. To be added here soon! □

This last result establishes the upper-semicontinuity of the spectrum. Loosely speaking, it tells us that an infinitesimal perturbation of T can enlarge $\sigma(T)$ only infinitesimally [3].

2.4 The Spectral Radius of Bounded Operators

Next, we consider the spectral radius of a bounded linear operator, defined as follows:

Definition 2.7 (Spectral radius [4]). *Let X be a Banach space and $T \in B(X, X)$. The spectral radius of T is the number*

$$r_T = \sup_{z \in \sigma(T)} |z|. \quad (14)$$

When considering the spectral radius of a bounded operator, we often make use of Gelfand's formula:

Theorem 2.8 (Gelfand's Formula). *Let X be a Banach space and $T \in B(X, X)$. The spectral radius of T can be computed from the following limit:*

$$r_T = \lim_{n \rightarrow \infty} \|T^n\|_X^{1/n}. \quad (15)$$

Using these two definitions of the spectral radius, we can prove the following results:

Theorem 2.9. *Let H be a Hilbert space and let $T \in B(H, H)$ be self-adjoint. Then $r_T = \|T\|_H$.*

Proof. Since T is self-adjoint, $\|T^2\|_H = \|T\|_H^2$ ⁶. Thus, $\|T^{2n}\|_H^{\frac{1}{2n}} = \|T\|_H$. Applying Gelfand's formula gives the result. $\square$

Theorem 2.10. *Let H be a Hilbert space and let $T \in B(H, H)$ be self-adjoint. If $\sigma(T) = \{0\}$ then $T = 0$.*

Proof. If $\sigma(T) = \{0\}$, then $r_T = \|T\|_H = 0$. Thus, $T = 0$. $\square$

2.5 Spectra of Compact Operators

In this section, we state and prove some properties of the spectra of compact linear operators. First, we prove that for compact linear operators on infinite-dimensional spaces, the only possible spectral value that is not an eigenvalue is zero.

Theorem 2.11 (Zero spectral value [4]). *Let X be an infinite-dimensional Banach space and $T \in K(X, X)$. Then $0 \in \sigma(T)$ and $\sigma(T) \setminus \{0\} = \Gamma(T) \setminus \{0\}$.*

Proof. In written notes. To be added here soon! $\square$

Next, we show that compact linear operators have isolated spectra.

Theorem 2.12 (Isolated spectra [4]). *Let X be a Banach space and $T \in K(X, X)$. Consider $\{z_n\}_{n \in \mathbb{N}}$ be a sequence in $\sigma(T) \setminus \{0\}$. If $z_n \rightarrow z$ then $z = 0$. Thus, $\sigma(T) \setminus \{0\}$ is made up of isolated points.*

Proof. In written notes. To be added here soon! $\square$

Using these results, the spectrum of a compact linear operator can be characterized as follows:

Corollary 2.13. *Let X be a Banach space and $T \in K(X, X)$. Exactly one of the following holds:*

1. $\sigma(T) = \{0\}$
2. $\sigma(T) \setminus \{0\}$ is finite.
3. $\sigma(T) \setminus \{0\}$ is a sequence converging to zero.

Proof. In written notes. To be added here soon! $\square$

⁶A proof of this result is in my written notes.

Using this characterization, we have the following result for a finite-rank bounded linear operator:

Corollary 2.14. *Let X be a Banach space and $T \in B(X, X)$ be a finite-rank operator, i.e., $\dim(R(T)) < \infty$. Then $\sigma(T) \setminus \{0\}$ has finitely many elements.*

Proof. In written notes. To be added here soon! □

2.6 Spectral Theorem for Normal and Self-Adjoint Operators

Recall that a Hilbert space is called separable if it has a dense countable subset. Every separable Hilbert space has a Hilbert basis, defined as follows:

Definition 2.15 (Hilbert basis [4]). *Let $\{e_n\}_{n \in \mathbb{N}}$ be a sequence of orthonormal elements of the Hilbert space H . Then, $\{e_n\}_{n \in \mathbb{N}}$ is called a Hilbert basis of H if $\overline{\text{span}\{e_n \mid n \in \mathbb{N}\}} = H$.*

Using this definition, we can state and prove the following result for compact and self-adjoint linear operators. We note that the same result holds for compact and normal operators [4].

Theorem 2.16. *Let H be a separable Hilbert space. Let $T \in K(H, H)$ be self-adjoint. Then, there exists a Hilbert basis composed of the eigenvectors of T .*

Proof. In written notes. To be added here soon! □

3 An Introduction to Pseudospectra - Coming Soon.

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